# CH 4 - Inner Product Spaces

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We defined abstract vector spaces to model $\mathbb{R}^n$.

So far, the focus has been on algebra. Geometry is rarely discussed.

In $\mathbb{R}^n$, we can talk about lengths, distance, and angles via DOT PRODUCT

Recall that in $\mathbb{R}^n$, $\vec{x} \cdot \vec{y} = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} = x_1 y_1 + \dots + x_n y_n$

**Observations**:

1. $\vec{x} \cdot \vec{x} = x_1^2 + \dots + x_n^2 = \|\vec{x}\|^2$
2. $\text{dist}(x, y) = \|\vec{y} - \vec{x}\| = \sqrt{(\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y})}$
3. $\vec{x} \cdot \vec{y} = \|x\|\|y\| \cos\theta \implies \theta = \arccos \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$

We can view dot product as $f : V \times V \to \mathbb{R}$, $f(x, y) = x \cdot y \in \mathbb{R}$ that satisfies

1. $f(x, y) = f(y, x)$, symmetry
2. $f(cx + z, y) = cf(x, y) + f(z, y)$, linearity in the first argument(second argument is same by symmetry)
3. $f(x, x) \geq 0 \ \forall x$, $f(x, x) = 0 \iff x = 0$, positive-definite
4. $\frac{f(x,y)}{\|x\|\|y\|} \in [-1, 1]$, can be obtained from the above 3 properties.

> 📘 **Definition — Inner Product**
>
> Let $V$ be a vector space over $\mathbb{F}$. An **inner product** on $V$ is a function
>
> $$\langle\ ,\ \rangle : V \times V \to \mathbb{F}$$
>
> satisfying:
>
> 1. Symmetry: $\langle \vec{x}, \vec{y} \rangle = \overline{\langle \vec{y}, \vec{x} \rangle}$
> 2. Linearity in arguments: $\langle a\vec{x} + y, \vec{z} \rangle = a\langle \vec{x}, \vec{z} \rangle + \langle \vec{y}, \vec{z} \rangle$
> 3. Positive-definite: $\langle \vec{x}, \vec{x} \rangle \geq 0$ and $\langle \vec{x}, \vec{x} \rangle = 0 \iff \vec{x} = \vec{0}$

Examples

1. $V = \mathbb{R}^n$, $\langle \vec{x}, \vec{y} \rangle = \vec{x} \cdot \vec{y}$ is an inner product
2. $V = P_2(\mathbb{R})$, $\langle a_0 + a_1 x + a_2 x^2, b_0 + b_1 x + b_2 x^2 \rangle = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = a_0 b_0 + a_1 b_1 + a_2 b_2$
3. $[L^2 \text{ inner product}] V = P_n(\mathbb{R})$

   $\langle p(x), q(x) \rangle = \int_0^1 p(x) q(x)\, \mathrm{d}x$

   Claim: the above is an inner product on $P_{n(\mathbb{R})}$

   1. Symmetry: $\langle p, q \rangle = \int_0^1 p(x) q(x)\, \mathrm{d}x = \int_0^1 q(x) p)(x)\, \mathrm{d}x = \langle q, p \rangle$

2. Linearity $\langle ap(x)+q(x), r(x)\rangle = \int_0^1 (ap(x)+q(x))r(x)\,\mathrm{d}x = a\int_0^1 p(x)r(x)\,\mathrm{d}x + \int_0^1 q(x)r(x)\,\mathrm{d}x = a\langle p(x), r(x)\rangle + \langle q(x), r(x)\rangle$
3. Absolute-definite $\langle p(x), p(x)\rangle = \int_0^1 p(x)p(x)\,\mathrm{d}x = \int_0^1 p^2(x)\,\mathrm{d}x \geq 0$

   If $\langle p(x), p(x)\rangle = 0$, then $\int_0^1 p^2(x)\,\mathrm{d}x = 0 \Longrightarrow p(x) = 0$

4. $V = M_{2\times 2}(\mathbb{R})$

   $\langle \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \begin{bmatrix} x & y \\ z & w \end{bmatrix} \rangle = ax + by + cz + dw$

   Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$

   $\langle A, B\rangle = \mathrm{tr}(B^T A)$ (Frobenius Inner Product on $M_{n\times n}(\mathbb{R})$)

   Special case, $M_{n\times 1} = \mathbb{R}^n$

   $\langle \vec{x}, \vec{y}\rangle_F = \mathrm{tr}(\vec{y}^T\vec{x}) = \mathrm{tr}([x_1y_1 + ... + x_ny_n]) = x_1y_1 + ... + x_ny_n = \vec{x}\cdot\vec{y}$

**Propositions of Inner Products**

1. By symmetry, we have linearity in second argument in $\mathbb{R}^n$: $\langle \vec{x}, a\vec{y} + \vec{z}\rangle = a\langle \vec{x}, \vec{y}\rangle + \langle \vec{x}, \vec{z}\rangle$
2. $\langle \vec{x}, \vec{0}\rangle = \langle \vec{0}, \vec{v}\rangle = 0$
3. Every finite-dimensional vector space $V$ has an inner product
4. Every non-zero vector space $V$ has infinitely many inner product.
5. $V = \mathbb{C}^n, \langle x, y\rangle = x_1\cdot\overline{y}_1 + ... + x_n\cdot\overline{y}_n$
6. The second arguments are conjugate linear $\langle \vec{x}, a\vec{y} + \vec{z}\rangle = \overline{a}\langle \vec{x}, \vec{y}\rangle = +\langle \vec{x}, \vec{z}\rangle$

Example:

Intuitively, $\langle \vec{x}, \vec{x}\rangle = \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} \cdot \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} = i^2 + 0^2 + ... + 0^2 = -1$

We define $\langle \begin{bmatrix} z_1 \\ ... \\ z_n \end{bmatrix}, \begin{bmatrix} w_1 \\ ... \\ w_n \end{bmatrix} \rangle = z_1\overline{w}_1 + z_2\overline{w}_2 + ... + z_n\overline{w}_n$

Then, $\langle \vec{x}, \vec{x}\rangle = \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} \cdot \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} = i\cdot\overline{i} + ... + 0\cdot\overline{0} = -(-1) = 1$ satisfies absolute-definite

**Proof**

2. $\langle \vec{x}, \vec{0}\rangle = \langle \vec{x}, \vec{0} + \vec{0}\rangle = \langle \vec{x}, \vec{0}\rangle + \langle \vec{x}, \vec{0}\rangle - \langle \vec{x}, \vec{0}\rangle = \langle \vec{x}, \vec{0}\rangle - \langle \vec{x}, \vec{0}\rangle = 0$
3. $V \cong \mathbb{R}^n$

   $\langle x, y\rangle = [x]_B \cdot [y]_B$ is an inner product
4. There exists an inner product $\langle , \rangle$

   Consider $c\langle , \rangle,\ c \in \mathbb{R}_{>0}$

   There are infinitely many inner product

Recall:

1. $\vec{x} \cdot \vec{x} = x_1^2 + \ldots + x_n^2 = \|\vec{x}\|^2$
2. $\text{dist}(x, y) = \|\vec{y} - \vec{x}\| = \sqrt{(\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y})}$
3. $\vec{x} \cdot \vec{y} = \|x\|\|y\| \cos\theta \Longrightarrow \theta = \arccos \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$

**Definition — Norms, Distance, Angles**

Let $V$ be an inner product space(a vector space with a fixed choice of inner product)

The **norm** of $\vec{x} \in V$ is $\|\vec{x}\| = \sqrt{\langle \vec{x}, \vec{x} \rangle}$ is a non-negative real number

The **distance** between $\vec{x}, \vec{y} \in V$ is $\text{dist}(\vec{x}, \vec{y}) = \sqrt{\langle \vec{x}, \vec{y} \rangle}$ is non-negative real number

**IMPORTANT: DO NOT FORGET THE SQUARE ROOT**

Examples:

1. If $V = \mathbb{R}^n$, $\langle , \rangle$ be dot product. These definitions are the familiar ones from Euclidean Geomtery.
2. $V = M_{2\times 2}(\mathbb{R})$, $\langle , \rangle$ be the Frobenius inner product
   $\| \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \| = \sqrt{\langle A, A \rangle} = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{30}$
3. $V = P_2(\mathbb{R})$, $\langle p, q \rangle = \int_0^1 p(x)q(x)\,\mathrm{d}x$
   $\text{dist}(1 - x^2, 2 + 3x + 2x^2) = \sqrt{\langle -x^2, 2 + 3x + 2x^2 \rangle}$
   $= \sqrt{\int_0^1 (1 - x^2)(2 + 3x + 2x^2)\,\mathrm{d}x} = \sqrt{\frac{153}{10}}$

**Theorem — Properties of Norm**

Let $V$ be an inner product space

1. $\|\vec{x}\| \geq 0$ and $\|\vec{x}\| = 0 \iff \vec{x} = \vec{0}$ (Positive-definite)
2. $\|c\vec{x}\| = |c|\ \|\vec{x}\|$ (Homogeneity)
3. $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ (Triangle Inequality)

**Proof**

2. $\|c\vec{x}\| = \sqrt{\langle c\vec{x}, c\vec{x} \rangle} = \sqrt{c\bar{c}\langle \vec{x}, \vec{x} \rangle} = \sqrt{|c|^2\ \langle \vec{x}, \vec{x} \rangle} = |c|\ \sqrt{\langle \vec{x}, \vec{x} \rangle} = |c|\ \|\vec{x}\|, \quad c \in \mathbb{F}$
3. $$\begin{aligned} \|\vec{x} + \vec{y}\|^2 &= \langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle + \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{x} \rangle + \langle \vec{y}, \vec{y} \rangle \\ &= \|\vec{x}\|^2 + \langle \vec{x}, \vec{y} \rangle + \overline{\langle \vec{x}, \vec{y} \rangle} + \|\vec{y}\|^2 \\ &= \|\vec{x}\|^2 + 2\ \text{Re}(\langle \vec{x}, \vec{y} \rangle) + \|\vec{y}\|^2 \\ &\leq \|\vec{x}\|^2 + 2|\langle \vec{x}, \vec{y} \rangle| + \|\vec{y}\|^2 = \|\vec{x}\|^2 + 2\|\vec{x}\|\ \|\vec{y}\| + \|\vec{y}\|^2 = (\|\vec{x}\| + \|\vec{y}\|)^2 \end{aligned}$$

   $\Longrightarrow \|\vec{x} + \vec{y}\| \leq \|\vec{x} + \vec{y}\|$

□

**Theorem — Cauchy-Schwarz Inequality**

Let $V$ be an inner product space. Then

$$|\langle \vec{v}, \vec{w} \rangle| \leq \|\vec{v}\| \|\vec{w}\|, \ \forall \vec{v}, \vec{w} \in V$$

with equality if and only if $\vec{v}$ and $\vec{w}$ are scalar multiples of each other

Basic idea: if $\langle , \rangle$ is dot product on $\mathbb{R}^n \leq 1$

$|\langle x, y \rangle| = |x \cdot y| = \|x\| \ \|y\| \ |\cos\theta| \leq \|x\| \ \|y\|$

**Definition —**

If $x, y \in V$ are non-zero, let the angle between them to be $\theta = \arccos\left(\frac{\langle x,y \rangle}{\|x\|\|y\|}\right)$

$$\frac{\langle x, y \rangle}{\|x\|\|y\|} \leq 1$$

by Cauchy-Schwarz

Question: when $\theta = 90$?

Ans: $\frac{\langle x,y \rangle}{\|x\|\|y\|} = \cos 90 = 0 \implies \langle x, y \rangle = 0$

**Definition — Orthogonal in Inner Product Space**

Let $V$ be an inner product space. We say $\vec{x}, \vec{y} \in V$ are orthogonal, denoted

$$\vec{x} \perp \vec{y} \iff \langle \vec{x}, \vec{y} \rangle = 0$$

Examples:

1. $V \in \mathbb{R}^2, \langle , \rangle =$ dot product. $e_1 \perp e_2 \iff \langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rangle = 0$
2. $V = P_2(\mathbb{R}), \langle p, q \rangle = \int_0^1 p(x)q(x) \, \mathrm{d}x$

   $p = 1, q = x$ are not orthogonal b/c $\langle p, q \rangle = \langle 1, x \rangle = \int_0^1 x \, \mathrm{d}x = \frac{1}{2} \neq 0$

   If we define $\langle p, q \rangle = \int_{-1}^1 pq \, \mathrm{d}x$

   Then $1 \perp x \iff \langle 1, x \rangle = \int_{-1}^1 x \, \mathrm{d}x = 0$

**Theorem — Pythagorean Theorem**

Let $V$ be an inner product space.

If

$$x \perp y \implies \|\vec{x} + \vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2$$

**Proof**

By definition, $\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle = \langle x, x \rangle + 0 + 0 + \langle y, y \rangle = \|x\|^2 + \|y\|^2$ since $x \perp y$

□

**Definition — Generalized Orthogonal set and Pythagorean Theorem**

We say $S = \{v_1, ..., v_k\}$ is an orthogonal set if $v_i \perp v_j, \forall i \neq j$

If $S = \{v_1, ..., v_k\}$ is orthogonal, then $\|(v_1 + ...) + v_k\|^2 = \|v_1\|^2 + ... + \|v_k\|^2$

**IMPORTANT**: Suppose $S = \{v_1, ..., v_k\}$ is orthogonal and $v_i \neq \vec{0}$, then $S$ is linearly independent.

**Proof**

Consider the equation $c_1 v_1 + ... + c_k v_k = 0, v_i \in S$

Take the equation and take the inner product on both sides by $v_i$

$$\begin{aligned} \langle c_1 v_1 + ... + c_k v_k, v_i \rangle &= \langle 0, v_i \rangle \\ c_1\langle v_1, v_i\rangle + ... + c_i\langle v_i, v_i\rangle + ... + c_k\langle v_k, v_i\rangle &= 0 \\ c_i\langle v_i, v_i\rangle &= 0 \end{aligned}$$

$v_i \neq 0 \implies \langle v_i, v_i\rangle \neq 0 \implies c_i = 0$, so $S$ is linearly independent

□

**Definition — Orthogonal Basis**

An **orthogonal basis** for $V$ is a basis that is an orthogonal set.

**IMPORTANT**: Coordinates with orthogonal are very easy to find

Examples:

1. Let $V = \mathbb{R}^2; \rangle, \langle =$ dot product, $S = \{e_1, e_2\}$ is an orthogonal basis
2. $B = \left\{v_1 = \begin{bmatrix}1\\1\end{bmatrix}, v_2 \begin{bmatrix}-1\\1\end{bmatrix}\right\}$ is an orthogonal basis.

   Let $x = \begin{bmatrix}a\\b\end{bmatrix} \in \mathbb{R}^2$.

   $c_1, c_2 \in \mathbb{R}$ so that $x = c_1\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}-1\\1\end{bmatrix}$

   Take inner product on both sides with $v_1$

   $\langle x, v_1\rangle = \langle c_1 v_1 + c_2 v_2, v_1\rangle = c\langle v_1, v_1\rangle + c_2(0) \implies c_1 = \frac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle} = \frac{\begin{bmatrix}a\\b\end{bmatrix}\cdot\begin{bmatrix}1\\1\end{bmatrix}}{\begin{bmatrix}1\\1\end{bmatrix}\cdot\begin{bmatrix}1\\1\end{bmatrix}} = \frac{a+b}{2}$

   Similarily for $v_2 \implies c_2 = \frac{\langle x, v_2\rangle}{\langle v_2, v_2\rangle} = \frac{\begin{bmatrix}a\\b\end{bmatrix}\cdot\begin{bmatrix}-1\\1\end{bmatrix}}{\begin{bmatrix}-1\\1\end{bmatrix}\cdot\begin{bmatrix}-1\\1\end{bmatrix}} = -\frac{a+b}{2}$

   Verify: $\begin{bmatrix}a\\b\end{bmatrix} = \frac{a+b}{2}\begin{bmatrix}1\\1\end{bmatrix} - \frac{a+b}{2}\begin{bmatrix}-1\\1\end{bmatrix}$

**Definition — Projection**

Let $\vec{v}, \vec{w} \in V, \vec{w} \neq 0$. The **projection of $\vec{v}$ onto $\vec{w}$** is the vector

$$\text{proj}_{\vec{w}} \vec{v} = \frac{\langle \vec{v}, \vec{w} \rangle}{\langle \vec{w}, \vec{w} \rangle} \vec{w}$$

**Theorem — Coordinates with Orthogonal Basis**

Suppose $B = \{v_1, ..., v_n\}$ is an orthogonal basis for $V$. Then for all $x \in V$,

$$x = \sum_{i=1}^{n} \frac{\langle x, v_i \rangle}{\langle v_i, v_i \rangle} v_i = \sum_{i=1}^{n} \text{proj}_{\vec{v}_i} \vec{x}$$

**WARN**: This fails if $B$ is not orthogonal

Bad example:

$B = \left\{ v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ is a basis but not orthogonal

Take $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$\text{proj}_{v_1} x + \text{proj}_{v_2} x = \frac{\langle x, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 + \frac{\langle x, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 = \begin{bmatrix} \frac{7}{2} + \frac{11}{5} \\ \frac{7}{2} + \frac{22}{5} \end{bmatrix} \neq x$

**Definition — Perpendicular**

Let $\vec{w} \neq \vec{0}, \vec{v}$ be arbitrary. The perpendicular of $\vec{v}$ to $\vec{w}$ is

$$\text{perp}_{\vec{w}} \vec{v} = \vec{v} - \text{proj}_{\vec{w}} \vec{v}$$

1. $v = \text{proj}_{\vec{w}} \vec{v} + \text{perp}_{\vec{w}} \vec{v}$
2. $\text{proj}_{\vec{w}} \vec{v} \perp \text{perp}_{\vec{w}} \vec{v}$
3. $\text{proj}_{\vec{w}} \vec{v} \in \text{span}\{\vec{w}\}$

Question: Let $V$ be an inner product space. Does $V$ have an orthogonal basis?

Ans: Yes

If $\dim V = 0, B = \emptyset$ is orthogonal

If $\dim V = 1, B = \{v\}$ is orthogonal

If $\dim V = 2, B = \{v_1, v_2\}$, we can always fix $B$ to ba an orthogonal basis to $C = \{w_1, w_2\}$, take $w_1 = v_1$, then $w_2 = \text{perp}_{v_2} v_1$.

Additionally, $w_1 \perp w_2, \text{span}\{v_1, v_2\} = \text{span}\{w_1, w_2\}$

**Proof**

It is sufficient to show that $v_1, v_2 \in \text{span}\{w_1, w_2\}$

$v_1 = w_1, v_2 = v - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$

Similarily $w_1 = v_1, w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} v_1$

If dim $V = 3$?

dim $V = 3$

We have $B = \{v_1, v_2, v_3\}$, take $\begin{cases} w_1 = v_1 \\ w_2 = v_2 - \text{proj}_{v_1}(v_2) \\ w_3 = v_3 - \text{proj}_{v_1}(v_3) - \text{proj}_{w_2}(v_3) = v_3 - \text{proj}_W(v_3) \text{ where } W = \text{span}\{w_1, w_2\} \end{cases}$

Claim:

1. $\{w_1, w_2, w_3\}$ is an orthogonal set
2. $\text{span}\{w_1, w_2, w_3\} = \text{span}\{w_1, w_2, w_3\} = V$

Therefore, $\{w_1, w_2, w_3\}$ is an orthobasis for $V$

**Proof**

1. We want $\langle w_i, w_j \rangle = 0 \quad \forall i \neq j$

   We know that $\langle w_1, w_2 \rangle = 0$

   $w_3 = v_3 - \alpha w_1 - \beta w_2$

   $\langle w_3, w_1 \rangle = \langle v_3, v_1 \rangle - \alpha \langle v_1, v_1 \rangle - \beta \langle v_2, v_1 \rangle = \langle v_3, v_1 \rangle - \frac{\langle w_3, w_1 \rangle}{\langle w_1, w_1 \rangle} \langle w_1, w_1 \rangle = 0$

   $\langle w_3, w_2 \rangle = \langle v_3, v_2 \rangle - \alpha \langle v_2, v_1 \rangle - \beta \langle v_2, v_2 \rangle = \langle v_3, v_1 \rangle - \frac{\langle w_3, w_2 \rangle}{\langle w_2, w_2 \rangle} \langle w_2, w_2 \rangle = 0$

> 📘 **Definition — Gram-Schmidt Process**
>
> Suppose $\{v_1, v_2, ..., v_n\}$ is a basis for inner product space $V$.
>
> Define recursively $\{w_1, w_2, ..., w_n\}$ by
>
> $$\begin{aligned} w_1 &= v_1 \\ w_2 &= v_2 - \text{proj}_{w_1}(v_2) \\ w_3 &= v_3 - \text{proj}_{w_1}(v_3) - \text{proj}_{w_2}(v_3) \\ &... \\ w_n &= v_n - \text{proj}_{w_1}(v_n) - \text{proj}_{w_2}(v_n) - ... - \text{proj}_{w_{n-1}}(v_n) \end{aligned}$$
>
> Then $\{w_1, w_2, ..., w_n\}$ is an orthobasis for $V$
>
> **WARN: $w_i = v_i - \sum_{j=1}^{i} \text{proj}_{w_j}(v_i)$, the subscript is $w$ not $v$**
>
> After each step, we can scale the vector by a scalar to eliminate fractions.

Example:

Find an orthobasis for $V = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} \right\} \subseteq \mathbb{R}^4, \langle , \rangle = \text{dot product}$

$$w_1 = v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$w_2 = v_2 - \text{proj}_{w_1}(v_2) = \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} = 0$$

$$w_3 = v_3 - \text{proj}_{w_1}(v_3) - \text{proj}_{w_2}(v_3) = \frac{5}{6}\begin{bmatrix}-2\\1\\0\\1\end{bmatrix}, \frac{5}{6}\begin{bmatrix}-2\\1\\0\\1\end{bmatrix} \cdot \begin{bmatrix}1\\1\\0\\1\end{bmatrix} = \frac{5}{6}\begin{bmatrix}-2\\1\\0\\1\end{bmatrix} \cdot \begin{bmatrix}0\\1\\0\\-1\end{bmatrix} = 0$$

$$C = \left\{ \begin{bmatrix}1\\1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\\-1\end{bmatrix}, \begin{bmatrix}-2\\1\\0\\1\end{bmatrix} \right\} \text{ is an orthobasis}$$

**Note** that

1. Gram-Schmidt depends on the order of $v_1, ..., v_n$. A different ordering will produce different orthobasis
2. If there is a linear dependence relationship in $v_1, ..., v_n$. Then Gram-Schmidt will have the vector as $w_i = 0$

> 📘 **Definition — Orthonormal Basis**
>
> An **orthonormal basis** for $V$ is an orthogonal basis $B = \{v_1, ..., v_n\}$ for $V$ $\forall$ $\|v_i\| = 1$

> 🐋 **Info — Normalize Vector**
>
> Let $\vec{v}$ be a vector, a normalized $\vec{v}$ is
>
> $$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$

Example:

1. $C = \left\{ \begin{bmatrix}1\\1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\\-1\end{bmatrix}, \begin{bmatrix}-2\\1\\0\\1\end{bmatrix} \right\}$ is orthogonal, we can normalize it
   $$\hat{C} = \left\{ \frac{1}{\sqrt{3}}\begin{bmatrix}1\\1\\0\\1\end{bmatrix}, \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\0\\-1\end{bmatrix}, \frac{1}{\sqrt{6}}\begin{bmatrix}-2\\1\\0\\1\end{bmatrix} \right\}$$
2. Find an orthonormal basis for $P_2(\mathbb{R})$ with $\langle p, q\rangle = \int_0^1 p(x)q(x)\,\mathrm{d}x$

> 📐 **Theorem — Orthogonal Decomposition Theorem**
>
> Let $W$ be a subspace of the inner product space $V$. Let $\vec{v} \in V$. Then $\exists$ unique $\vec{p} \in W$ and $\vec{r} \perp W$, that is $\langle \vec{r}, \vec{w}\rangle = 0, \forall \vec{w} \in W$ s.t.
>
> $$\vec{v} = \vec{p} + \vec{r}$$
>
> Alternatively,
>
> If $W \subseteq V$ is a subpsace and if $\vec{v} \in V$, then
>
> $$\vec{v} = \text{proj}_W \vec{v} + \text{proj}_{W^\perp} \vec{v}$$

**Definition — Projection on Subspace**

Let $\vec{v} = \vec{p} + \vec{r}$ be as in the above theorem.

Then call $\vec{p}$ the **projection** of $\vec{v}$ onto $W$: $\vec{p} = \text{proj}_W \vec{v}$

Then call $\vec{r}$ the **perpendicular** of $\vec{v}$ onto $W$: $\vec{r} = \text{perp}_W \vec{v}$

The theorem says $\vec{v} = \text{proj}_W \vec{v} + \text{perp}_W \vec{v}$

Example:

1. The Orthogonal Decomposition Theorem looks like $\vec{v} = \text{proj}_{\vec{w}} \vec{v} + \text{perp}_{\vec{w}} \vec{v}$, where $\vec{w} \neq 0$

   If we let $W = \text{span}\{\vec{w}\}$, then we need to get $\text{proj}_W \vec{v} = \text{proj}_{\vec{w}} \vec{v}$ and $\text{perp}_W \vec{v} = \text{perp}_{\vec{w}} \vec{v}$

2. Take $V = \mathbb{R}^3$. $W = \text{xy-plane} = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R} \right\} = \text{span}\{e_1, e_{21}\}$

   Let $\vec{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in V$, then it is clear that $\text{proj}_W \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}, \text{perp}_W \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 7 \end{bmatrix}$

**Observation**: $\{e_1, e_2, e_3\}$ is an orthonormal basis for $\mathbb{R}^3$

$$\implies \vec{v} = \text{proj}_{e_1} \vec{v} + \text{proj}_{e_2} \vec{v} + \text{proj}_{e_2} \vec{v} = \frac{\langle v_1, e_1 \rangle}{\langle e_1, e_1 \rangle} e_1 + \frac{\langle v_1, e_2 \rangle}{\langle e_2, e_2 \rangle} e_2 + \frac{\langle v_1, e_3 \rangle}{\langle e_3, e_3 \rangle} e_3 = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ c \end{bmatrix}$$

**Question**: what is $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : 2x - y = x = 0 \right\}$?

Idea:

1. Find the basis for $W$
2. Apply Gram-Schmidt to the basis.
3. Extend to the basis for $V$
4. Apply Gram-Schmidt again

**ANS**:

1. Basis be $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \right\}$ by inspection
2. The basis is $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} \right\}$
3. $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ by extending the standard basis
4. Orthonormal basis $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$

Let $\{w_1, w_2, n\} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$

$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$\vec{p} = \text{proj}_{\vec{w}_1} \vec{v} + \text{proj}_{\vec{w}_2} \vec{v} = \frac{1}{2} \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$

$\vec{r} = \text{proj}_{\vec{n}}\vec{v} = \frac{1}{2}\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$

**Theorem — Lemma**

Let $W$ be a subspace, then $\exists$ an orthogonal basis of $V$ of the form $\{w_1, ..., w_k, n_1, ..., n_l\}$ where $\{w_1, ..., w_k\}$ is an orthobasis of $W$ and $n_1, ..., n_l \perp W$

**Proof**

Lemma

Let $\{v_1, ..., v_k\}$ be a basis for $W$ extend to a basis for $V = \{v_1, ..., v_k, y_1, ..., y_l\}$.

Apply Gram-Schmidt to the basis to get the orthobasis $\{w_1, ..., w_k, n_1, ..., n_l\}$. Then we know $\text{span}\{w_1, ..., w_k\} = \text{span}\{v_1, ..., v_k\} = W$

$\therefore \{w_1, ..., w_k\}$ is an orthobasis for $W$

Because linear independent as the outcome of Gram-Schmidt is already linearly independent since they are orthogonal and non-zero

It suffices to show that $n_i \perp w_j, \forall i, j$ because $w_j$ span $W$, b/c $\{w_1, ..., w_n, n_1, ..., n_l\}$ is an orthogonal set

□

**Proof**

Orthogonal Decomposition Theorem

Existence:

Choose an orthobasis $\{w_1, ..., w_k, n_1, ..., n_l\}$ like the lemma.

Given $\vec{v}_i$, we can write $\vec{v} = \sum_{i=1}^{k} \text{proj}_{w_i} v (\in W) + \sum_{j=1}^{l} \text{proj}_{n_i} v (\in W^{\perp})$

This is a formula for $\text{proj}_W \vec{v} + \text{perp}_W \vec{v}$

Uniqueness:

Let $v = p + r = \tilde{p} + \tilde{r}$ where $p, \tilde{p} \in W$ and $r, \tilde{r} \in W^{\perp}$
$\implies p + r = \tilde{p} + \tilde{r} \implies p - \tilde{p} (\in W) = r - \tilde{r} (\in W^{\perp})$
$\implies p - \tilde{p} \perp p - \tilde{p} \implies \langle p - \tilde{p}, p - \tilde{p} \rangle = 0 \iff p - \tilde{p} = 0 \iff p = \tilde{p}$
$\implies r - \tilde{r} \perp r - \tilde{r} \implies \langle r - \tilde{r}, r - \tilde{r} \rangle = 0 \iff r - \tilde{r} = 0 \iff r = \tilde{r}$ by positive definite.

□

**Definition — Proposition**

Let $V$ be an inner product space and $W \subseteq V$ be a subspace.

1. $W^{\perp} \in V$
2. $W \cup W^{\perp} = \left\{\vec{0}\right\}$ by positive definite
3. $\dim W + \dim W^{\perp} = \dim V$

Examples:

1. $S = \{\vec{0}\}, S^\perp = \{\vec{v} \in V : \vec{v} \perp \vec{0}\} = V$
2. $S = V, S^\perp = \{\vec{v} \in V : \vec{v} \perp V\} = \vec{0}$ by positive definite
3. $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : 2x - y + 5z = 0 \right\} \subseteq \mathbb{R}^3$ with dot product

   $W^\perp = \text{span}\left\{ \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} \right\}$
4. $W = \text{span}\left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\} \in M_{2\times 2}(\mathbb{R})$ with Frobenius inner product

   It suffices to find all $A \in V$ s.t. $\langle A, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \rangle = 0$ and $\langle A, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \rangle = 0$

   Let A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

   $\langle A, B \rangle = a + b + c + d = 0, \langle A, C \rangle = a + b = 0$

   $c = -d, a = -b \implies A = \begin{bmatrix} -b & b \\ -d & d \end{bmatrix} = b\begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}$

   $W^\perp = \text{span}\left\{ \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix} \right\}$

**Theorem — Shortest Distance Theorem**

Let $W \subseteq V$ is a subspace. Let $\vec{v} \in V$ have orghogonal decomposition

$$\vec{v} = \vec{p} + \vec{r}$$
$$\vec{p} = \text{proj}_W \vec{v}$$
$$\vec{r} = \text{proj}_{W^\perp} \vec{v}$$

Then $\vec{p}$ is the vector in $W$ closest to $\vec{v}$, i.e.

$$\text{dist}(\vec{v}, \vec{p}) \leq \text{dist}(\vec{v}, \vec{w}) \ \forall \vec{w} \in W$$
$$\|\vec{v} - \vec{p}\| \leq \|\vec{v} - \vec{w}\|$$

**Proof**

Shortest Distance Theorem

Consider a right triangle and let $v, p, w$ be vertices.

Then $\|v - w\|^2 = \|v - p + p - w\|^2 = \|(v - p) + (p - w)\|$ where $(v - p) \perp (p - w)$

$\implies \|v - p\|^2 + \|p - w\|^2 \geq \|v - p\|^2$

□

**Polynomial Approximation**

Example:

Let $V = \{\text{be continuous functions } f : [-1, 1] \to \mathbb{R}^2\}$

$W = \text{span}\{1, x, x^2, ..., x^N\}$

Let $f \in V$. Find a polynomial of degree at most $N$ that best approximates $f$.

We want $p \in W$ that is closest to $f = e^x$ and $p = \text{proj}_{W_N} f$

We need to choose the inner product: $\langle f, g \rangle = \int_{-1}^{1} fg$

We need an orthobasis, that is apply Gram-Schmidt to $W : \{1, x, x^2, ..., x^N\} \implies \{1, x, 3x^2 - 1, ...\}$

$\text{proj}_{W_1}(e^x) = \text{proj}_{w_0} e^x + \text{proj}_{w_1} e^x \approx 1.175 + 1.104x$

$\text{proj}_{W_2}(e^x) = \text{proj}_{w_0} e^x + \text{proj}_{w_1} e^x + \text{proj}_{w_2} e^x \approx 0.996 + 1.104x + 0.537x^2$

$\text{proj}_{W_3}(e^x) = \text{proj}_{w_0} e^x + \text{proj}_{w_1} e^x + \text{proj}_{w_2} e^x + \text{proj}_{w_3} e^x e^x \approx 0.996 + 0.998x + 0.537x^2 + 0.176x^3$

**TEST2 CUTOFF**